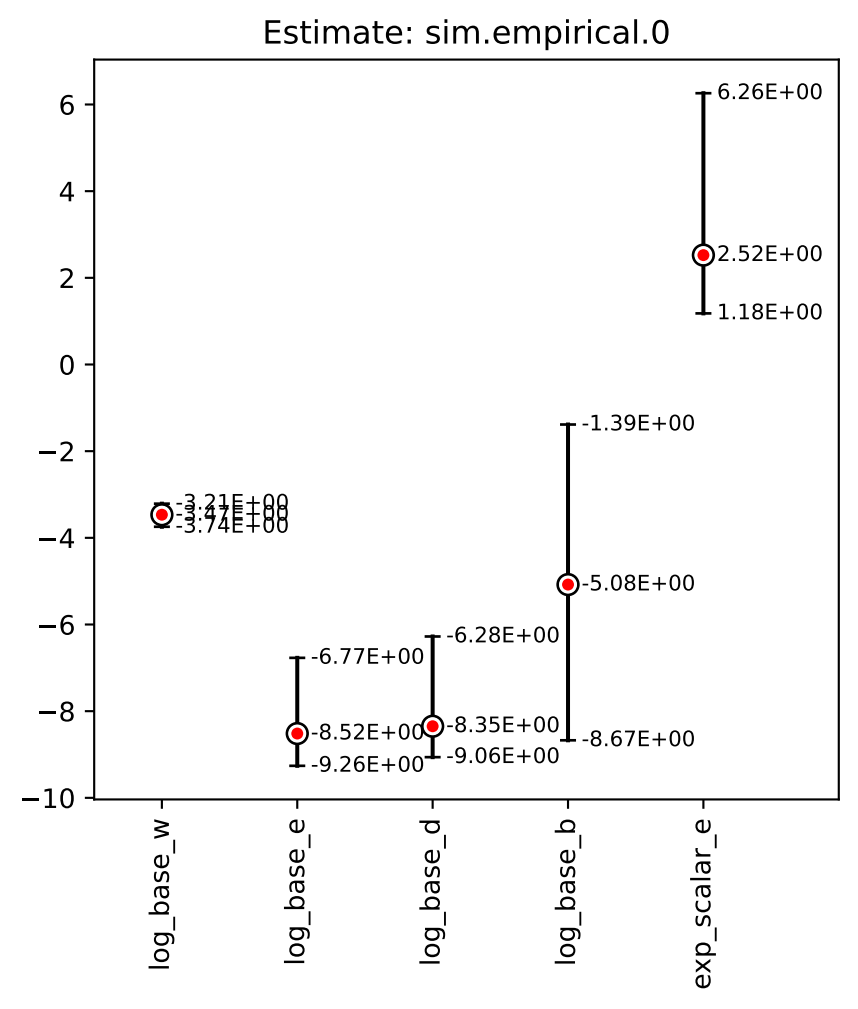
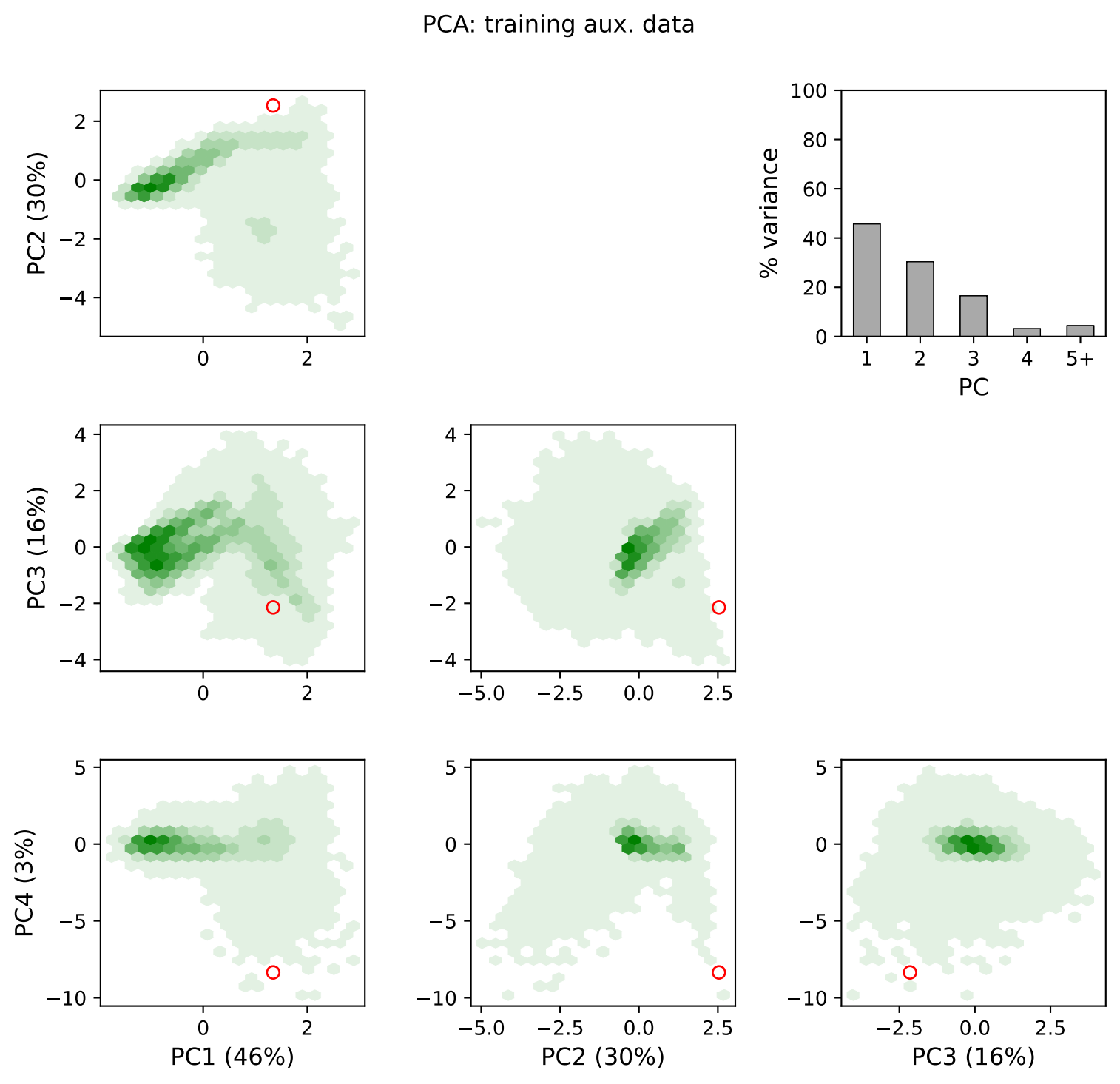
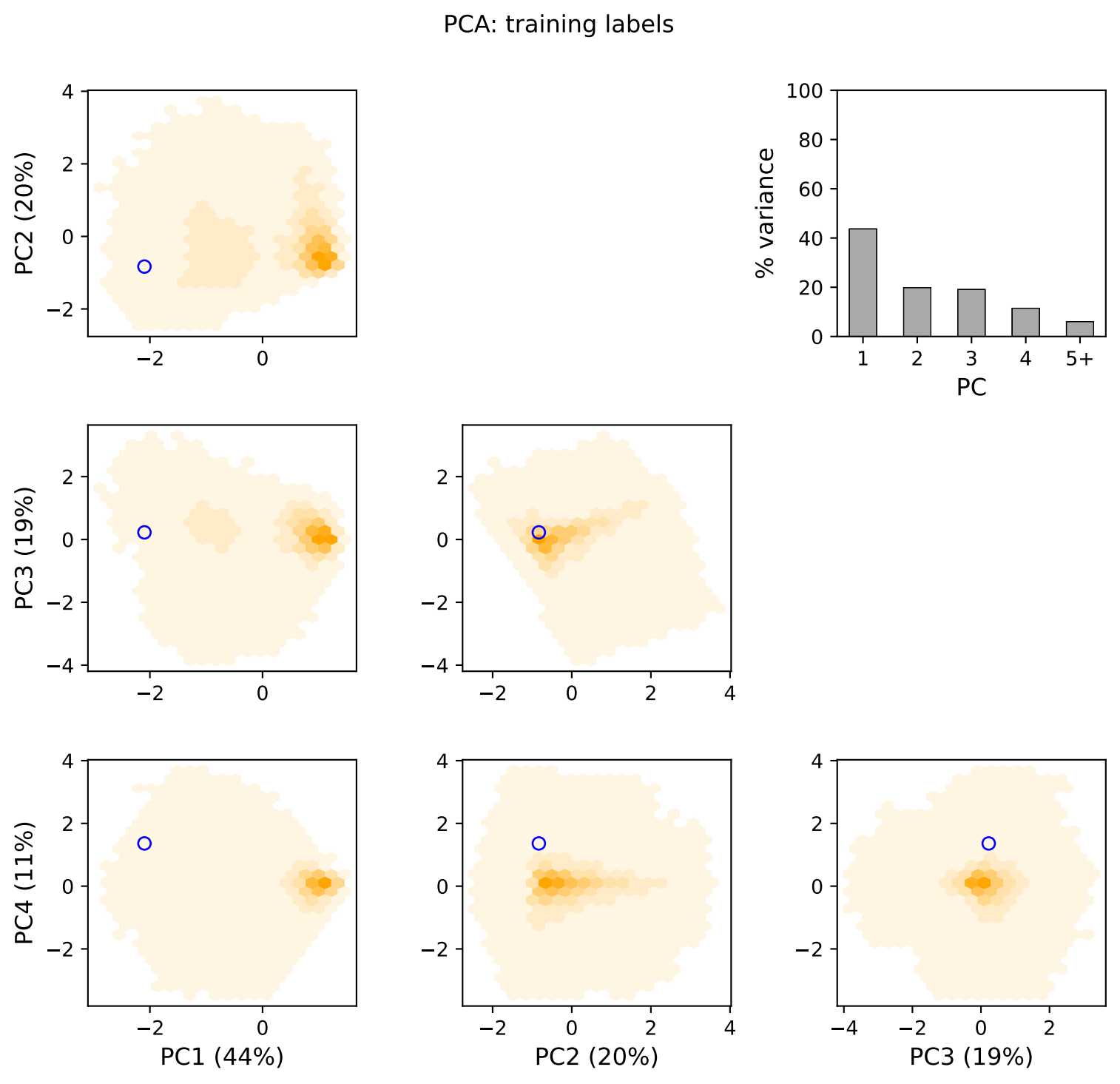
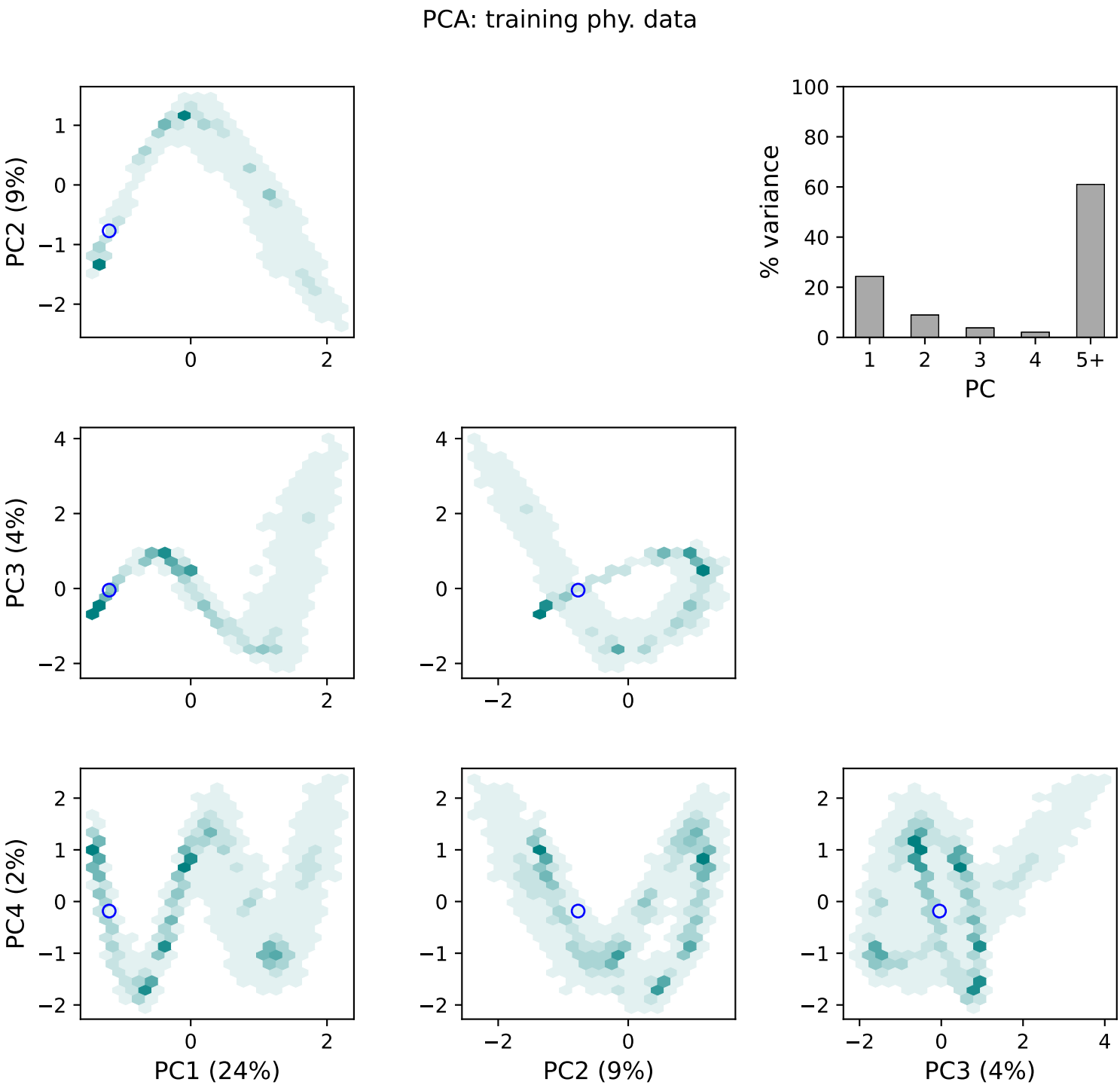
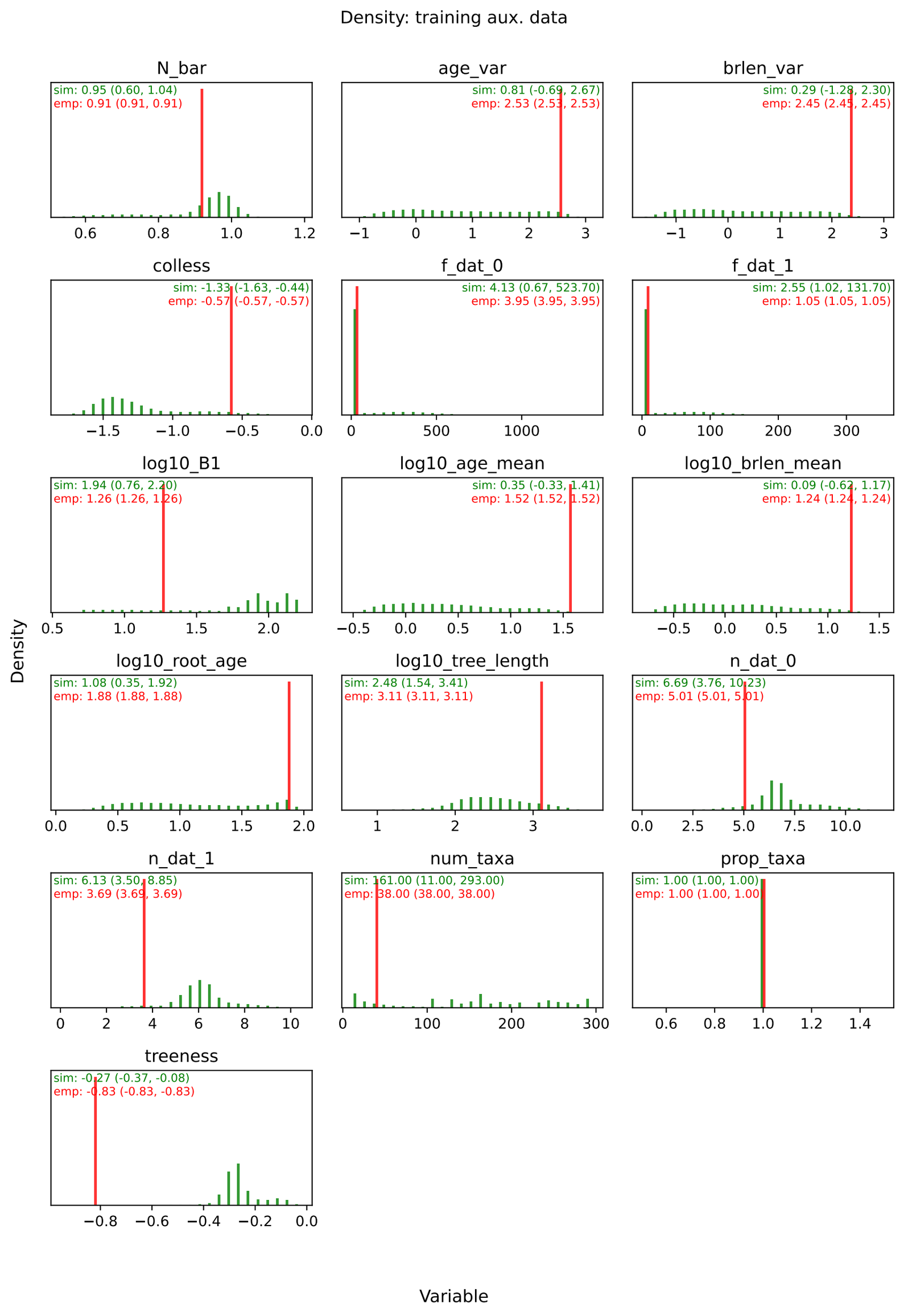
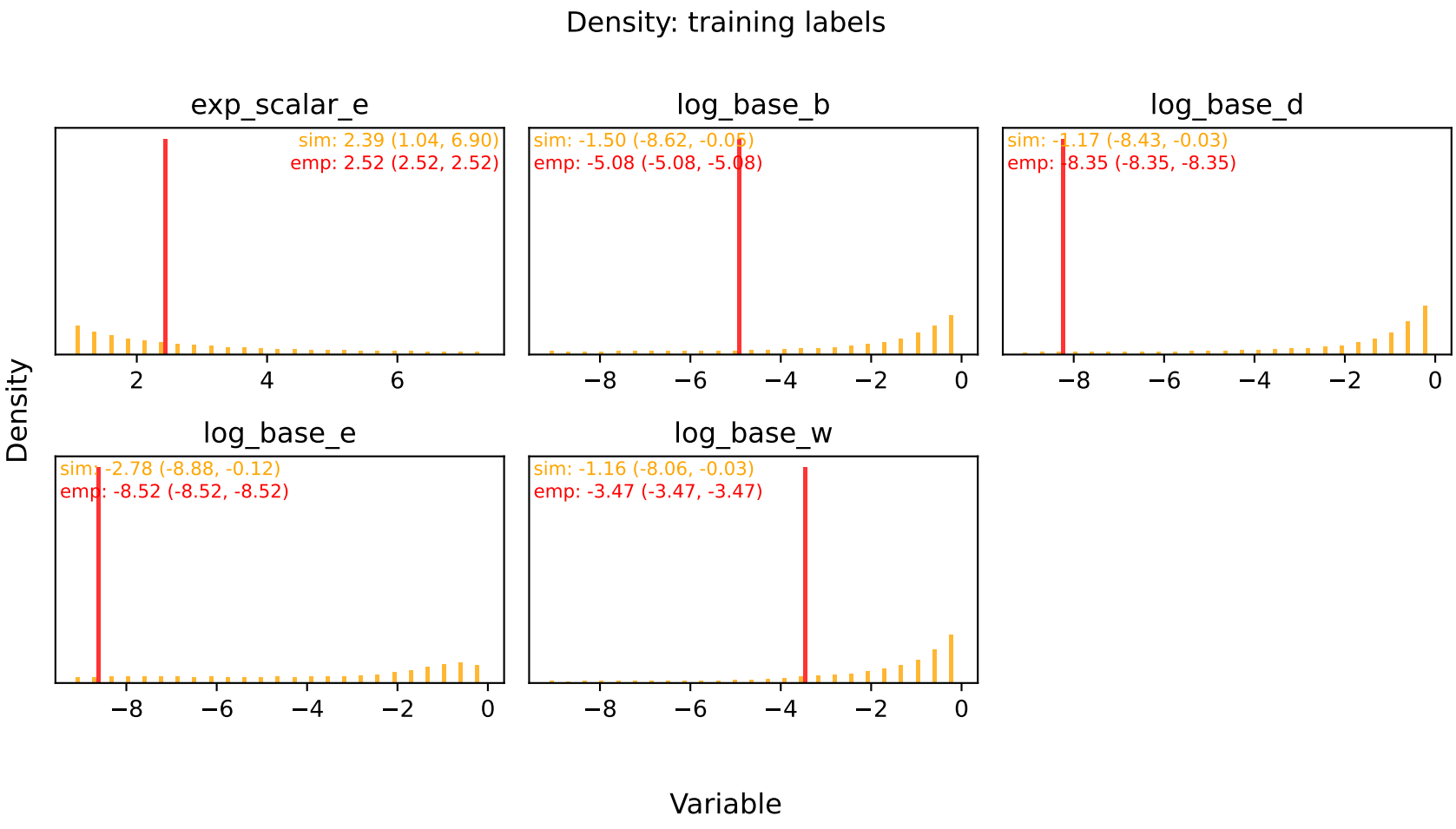

Estimate: sim.empirical.0
6
4
2
0
−2
−4
−6
−8
−10
6.26E+00
2.52E+00
1.18E+00
-1.39E+00
-5.08E+00
-8.67E+00
-3.21E+00
-3.47E+00
-3.74E+00
-6.77E+00
-8.52E+00
-9.26E+00
-6.28E+00
-8.35E+00
-9.06E+00
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_e

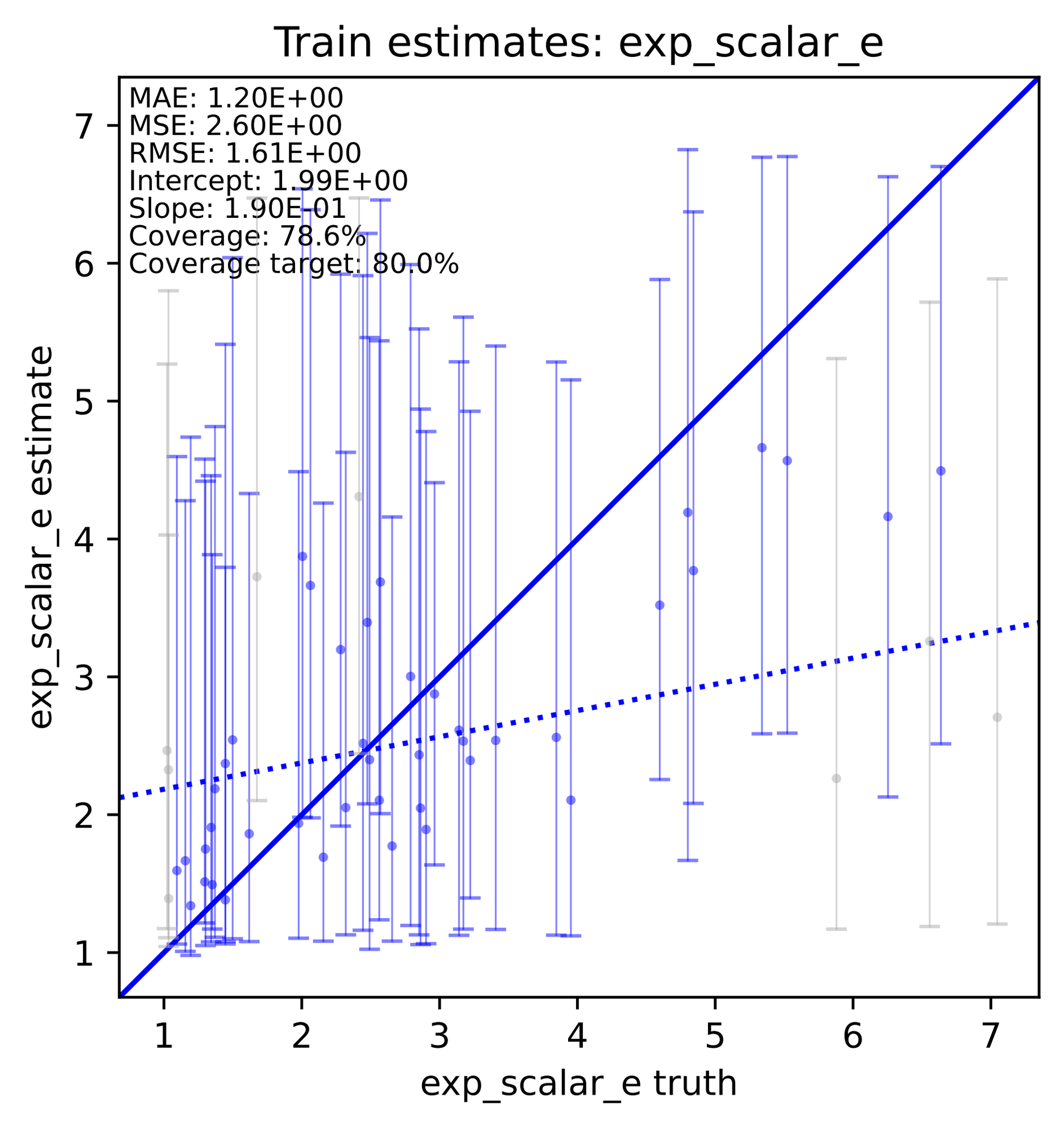

PCA: training aux. data
PC2 (30%)
% variance
PC
PC3 (16%)
PC4 (3%)
PC1 (46%)
PC2 (30%)
PC3 (16%)

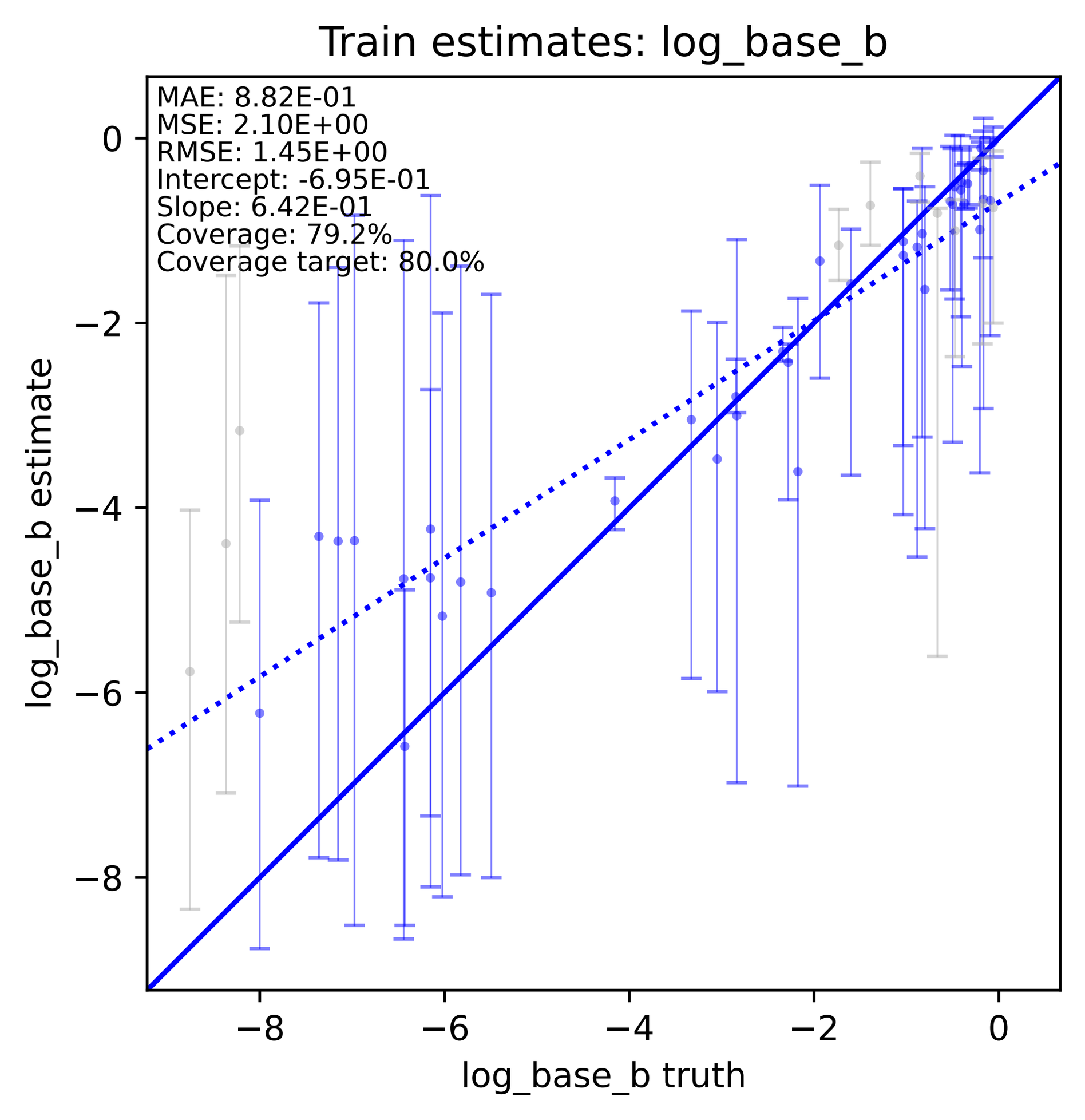

PCA: training labels
PC2 (20%)
% variance
PC
PC3 (19%)
PC4 (11%)
PC1 (44%)
PC2 (20%)
PC3 (19%)

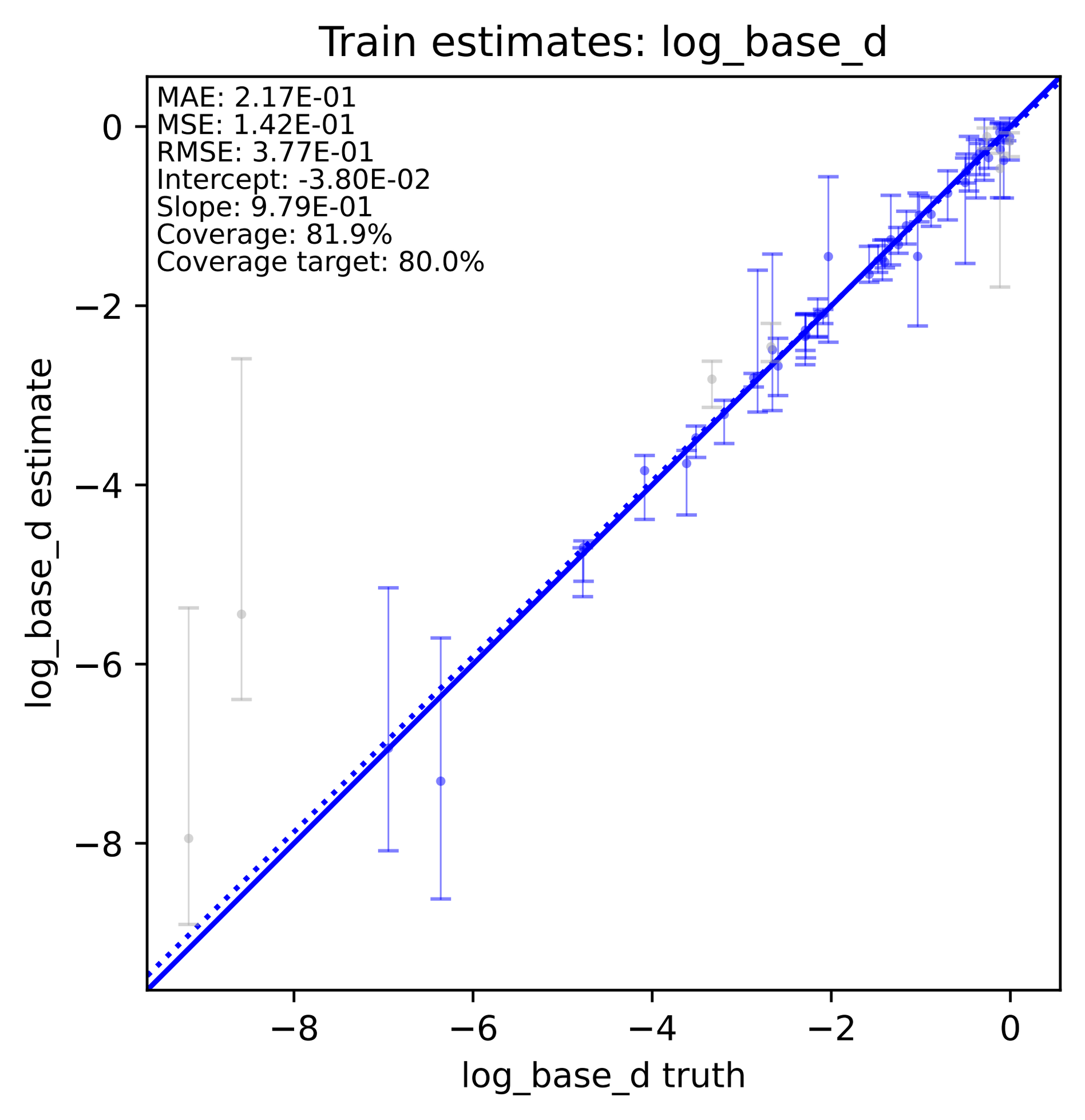

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (24%)
PC2 (9%)
PC3 (4%)

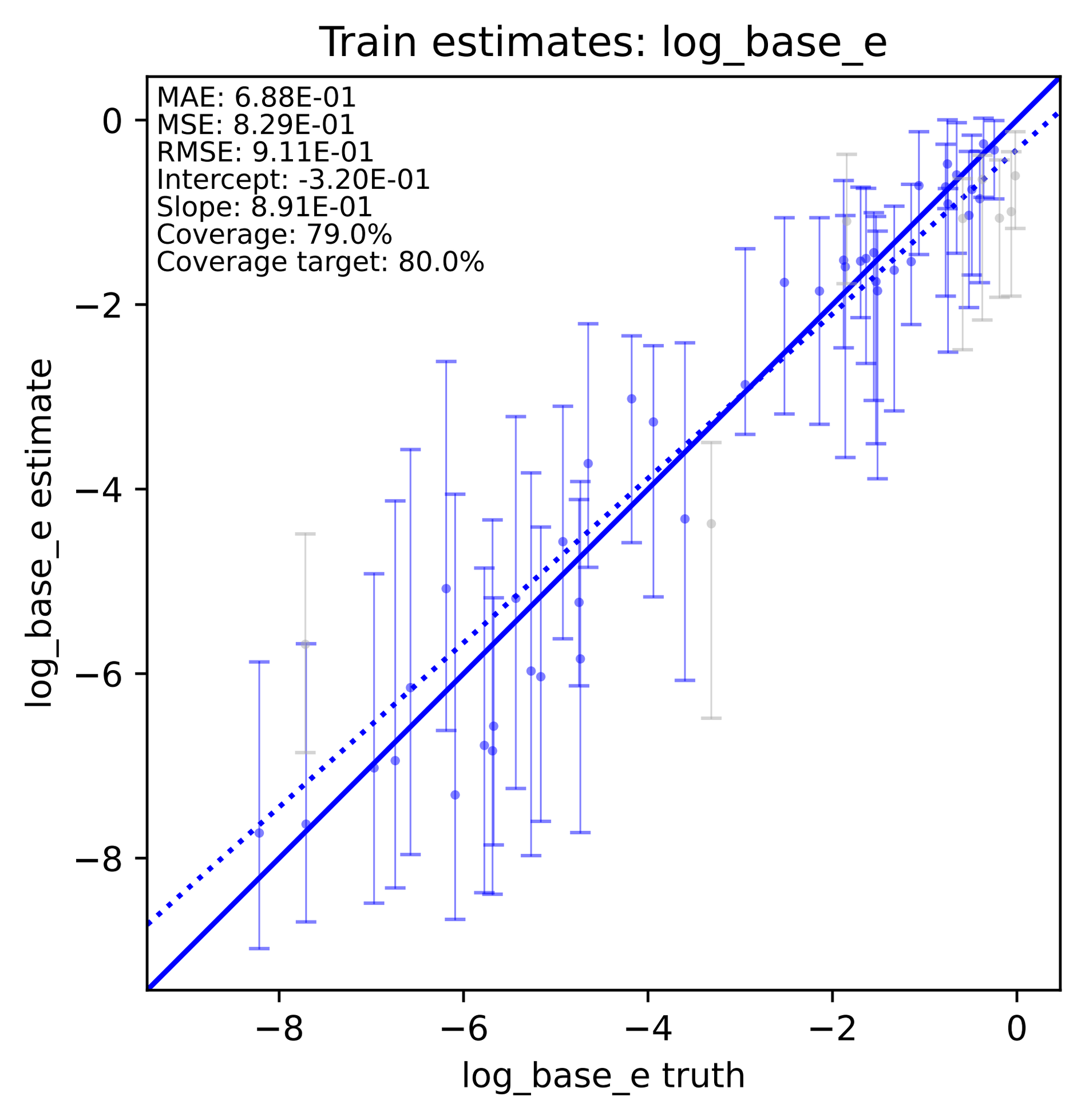

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 0.81 (-0.69, 2.67)
emp: 2.53 (2.53, 2.53)
brlen_var
sim: 0.29 (-1.28, 2.30)
emp: 2.45 (2.45, 2.45)
colless
sim: -1.33 (-1.63, -0.44)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 4.13 (0.67, 523.70)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.55 (1.02, 131.70)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 1.94 (0.76, 2.20)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.35 (-0.33, 1.41)
emp: 1.52 (1.52, 1.52)
log10_brlen_mean
sim: 0.09 (-0.62, 1.17)
emp: 1.24 (1.24, 1.24)
log10_root_age
sim: 1.08 (0.35, 1.92)
emp: 1.88 (1.88, 1.88)
log10_tree_length
sim: 2.48 (1.54, 3.41)
emp: 3.11 (3.11, 3.11)
n_dat_0
sim: 6.69 (3.76, 10.23)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 6.13 (3.50, 8.85)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 161.00 (11.00, 293.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.27 (-0.37, -0.08)
emp: -0.83 (-0.83, -0.83)
Density
Variable

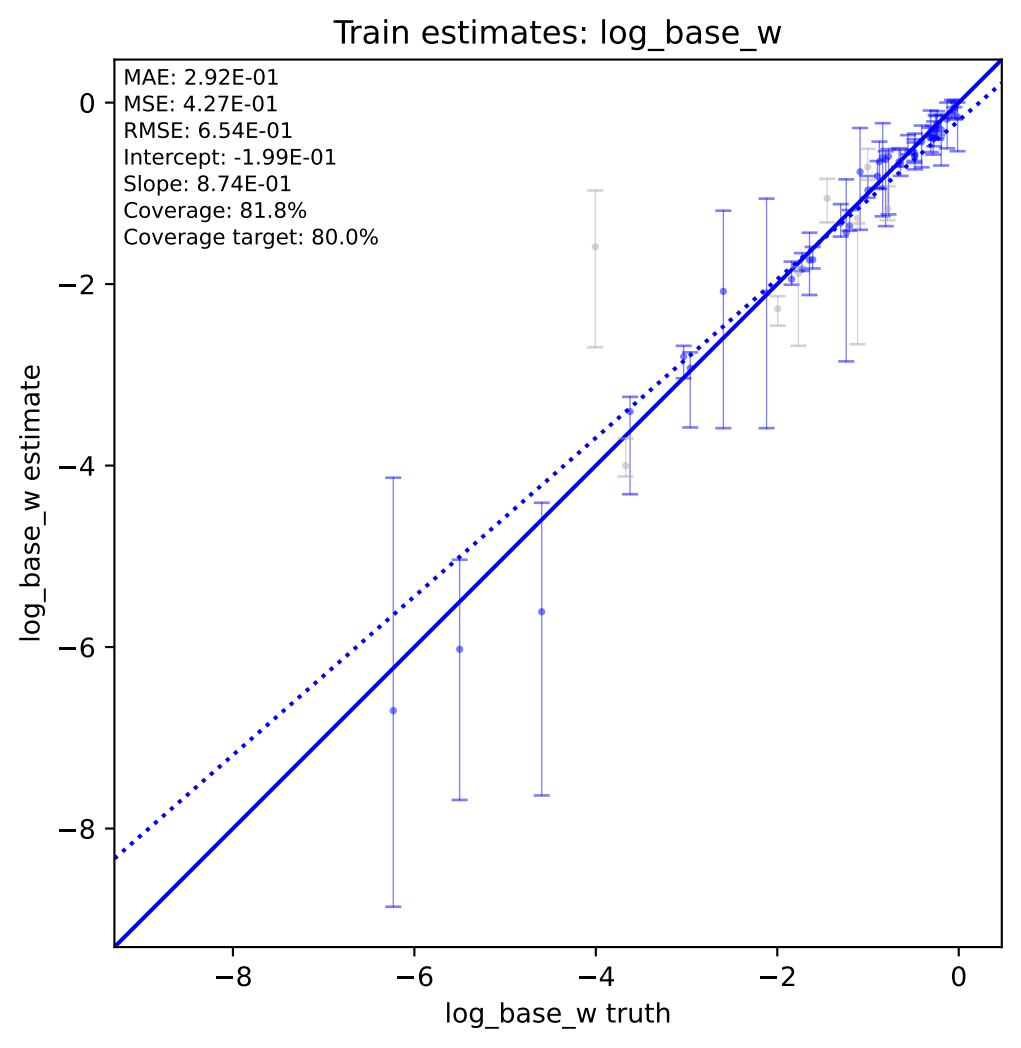

Density: training labels
exp_scalar_e
sim: 2.39 (1.04, 6.90)
emp: 2.52 (2.52, 2.52)
log_base_b
sim: -1.50 (-8.62, -0.05)
emp: -5.08 (-5.08, -5.08)
log_base_d
sim: -1.17 (-8.43, -0.03)
emp: -8.35 (-8.35, -8.35)
log_base_e
sim: -2.78 (-8.88, -0.12)
emp: -8.52 (-8.52, -8.52)
log_base_w
sim: -1.16 (-8.06, -0.03)
emp: -3.47 (-3.47, -3.47)
Density
Variable

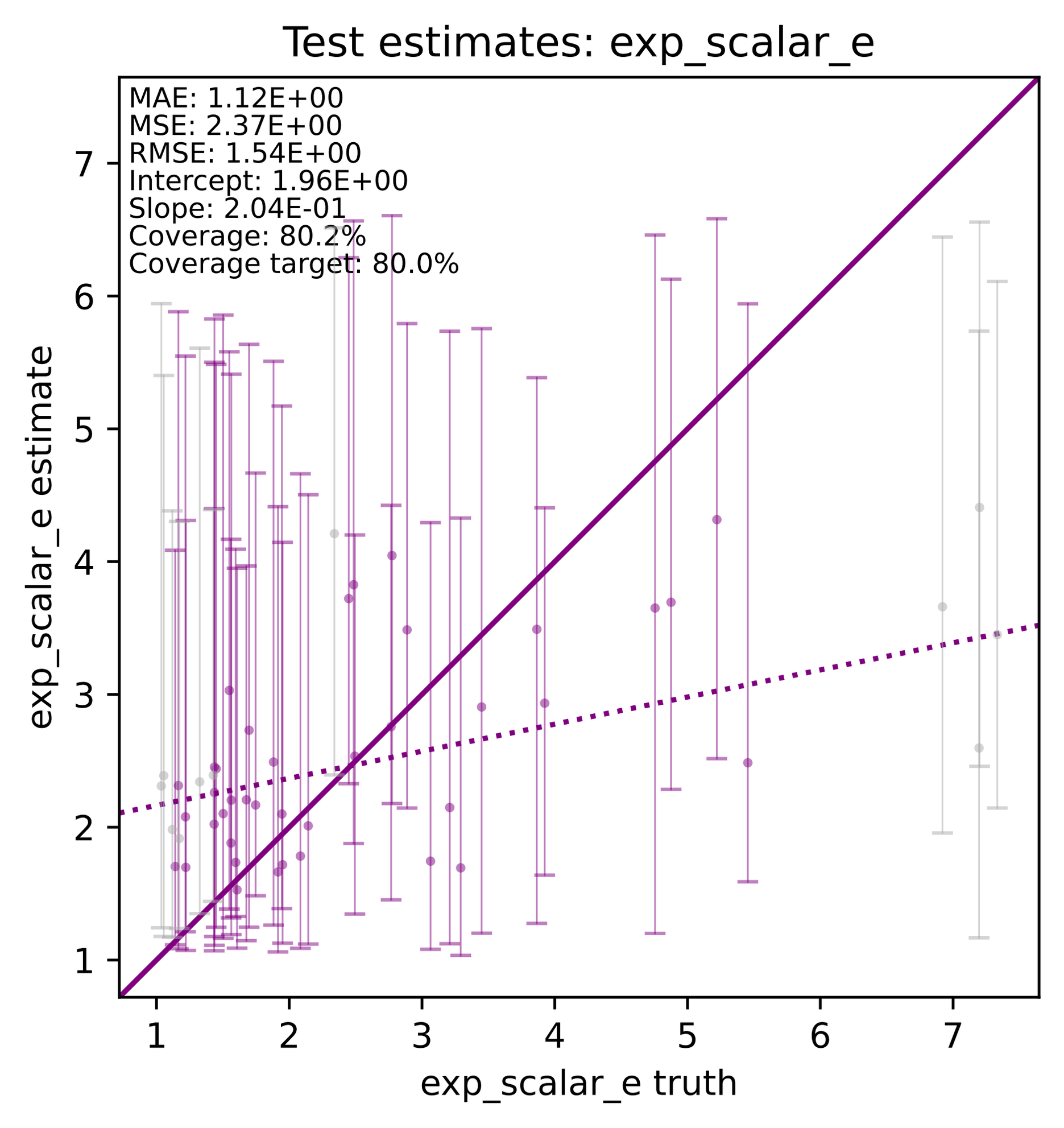

Train estimates: exp_scalar_e
MAE: 1.20E+00
MSE: 2.60E+00
RMSE: 1.61E+00
Intercept: 1.99E+00
Slope: 1.90E-01
Coverage: 78.6%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

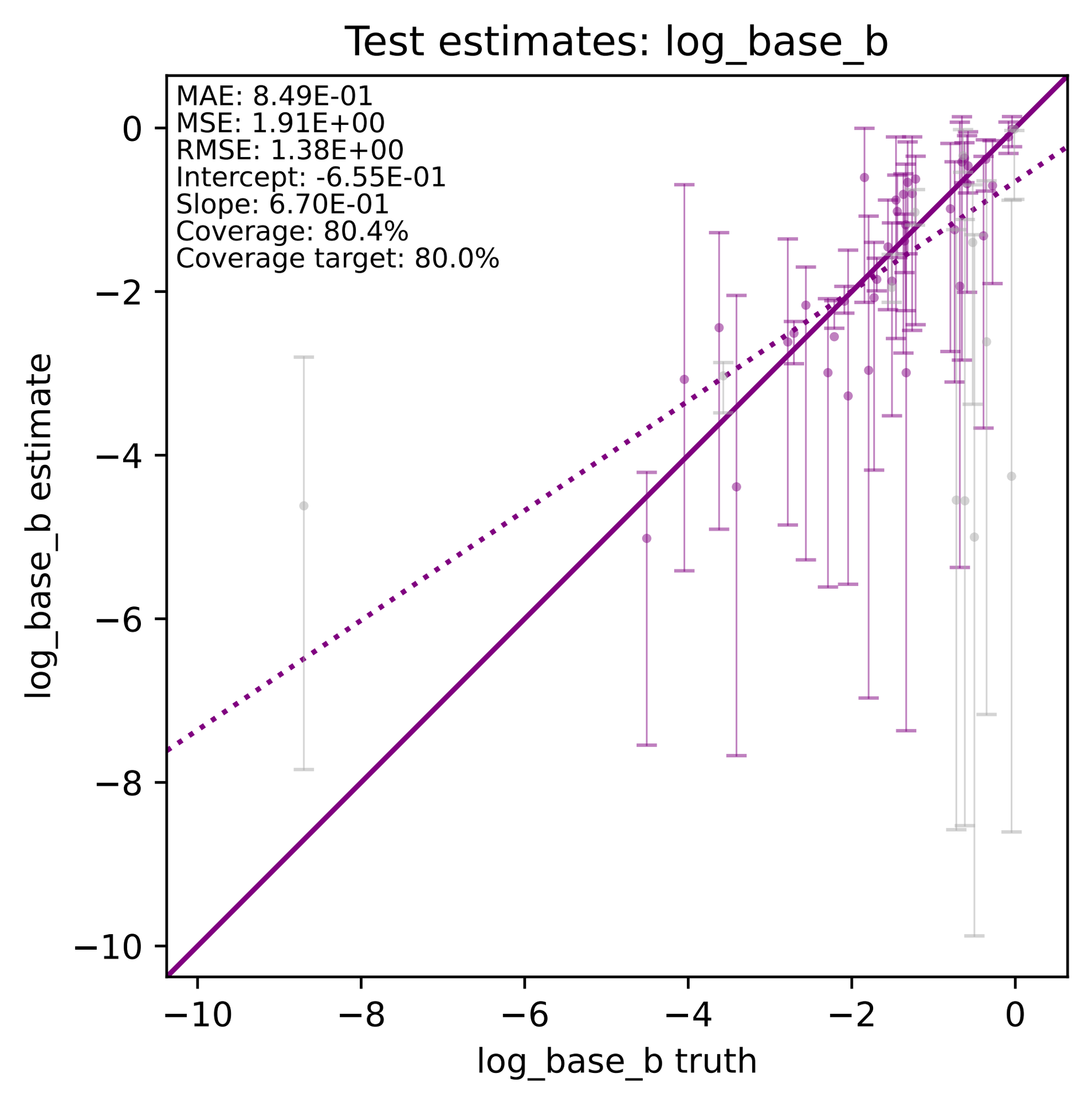

Train estimates: log_base_b
MAE: 8.82E-01
MSE: 2.10E+00
RMSE: 1.45E+00
Intercept: -6.95E-01
Slope: 6.42E-01
Coverage: 79.2%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

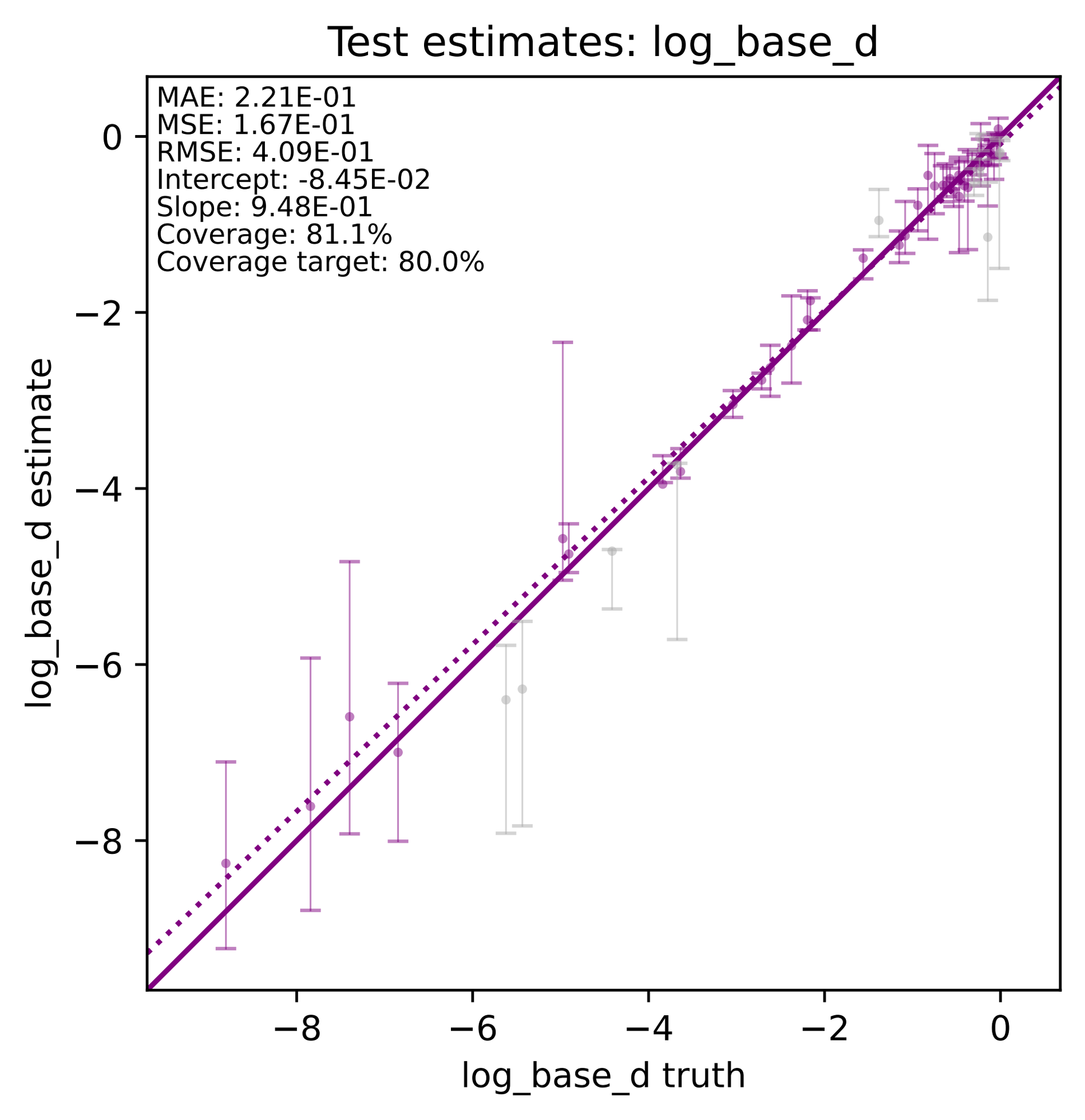

Train estimates: log_base_d
MAE: 2.17E-01
MSE: 1.42E-01
RMSE: 3.77E-01
Intercept: -3.80E-02
Slope: 9.79E-01
Coverage: 81.9%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

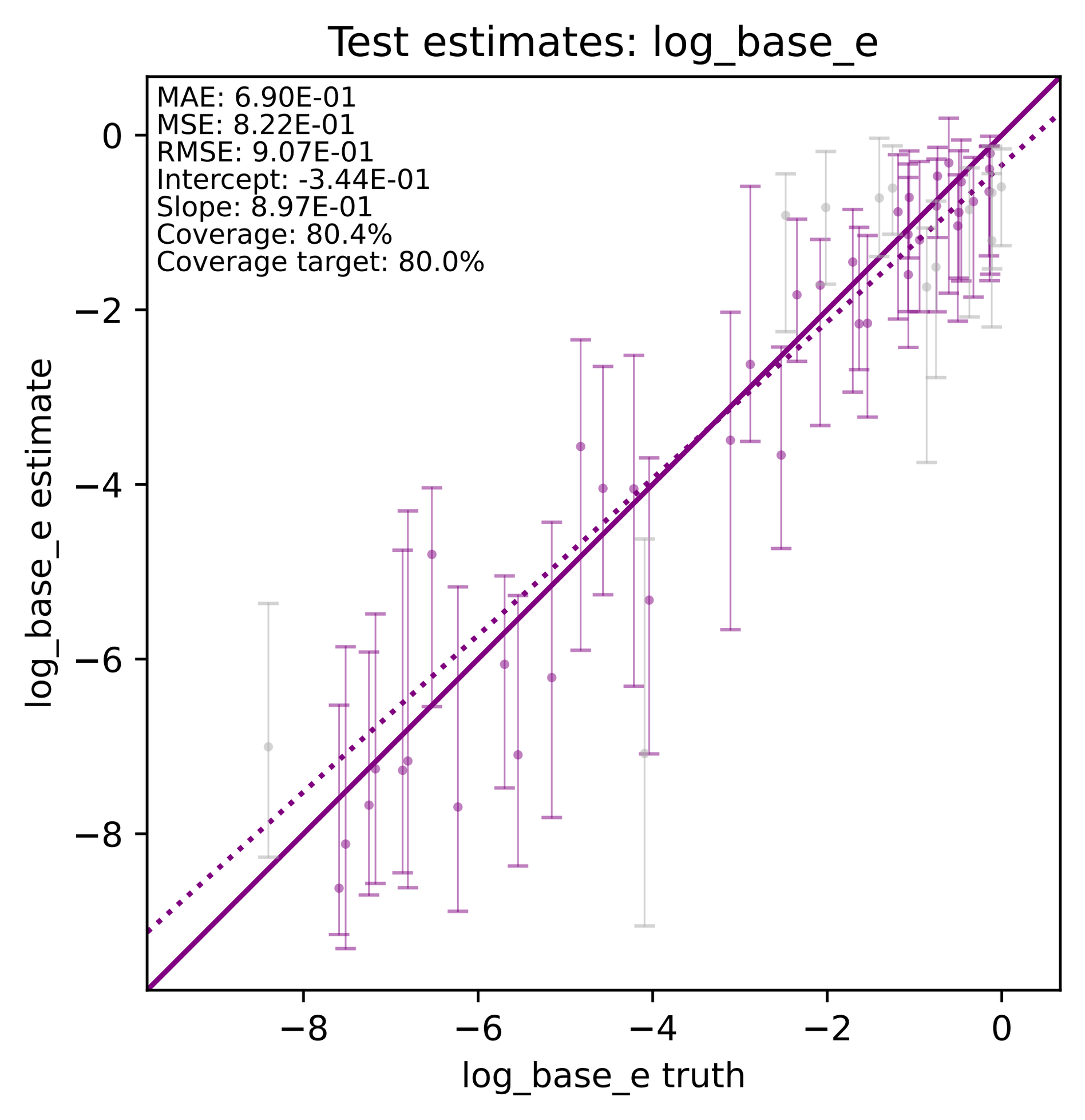

Train estimates: log_base_e
MAE: 6.88E-01
MSE: 8.29E-01
RMSE: 9.11E-01
Intercept: -3.20E-01
Slope: 8.91E-01
Coverage: 79.0%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

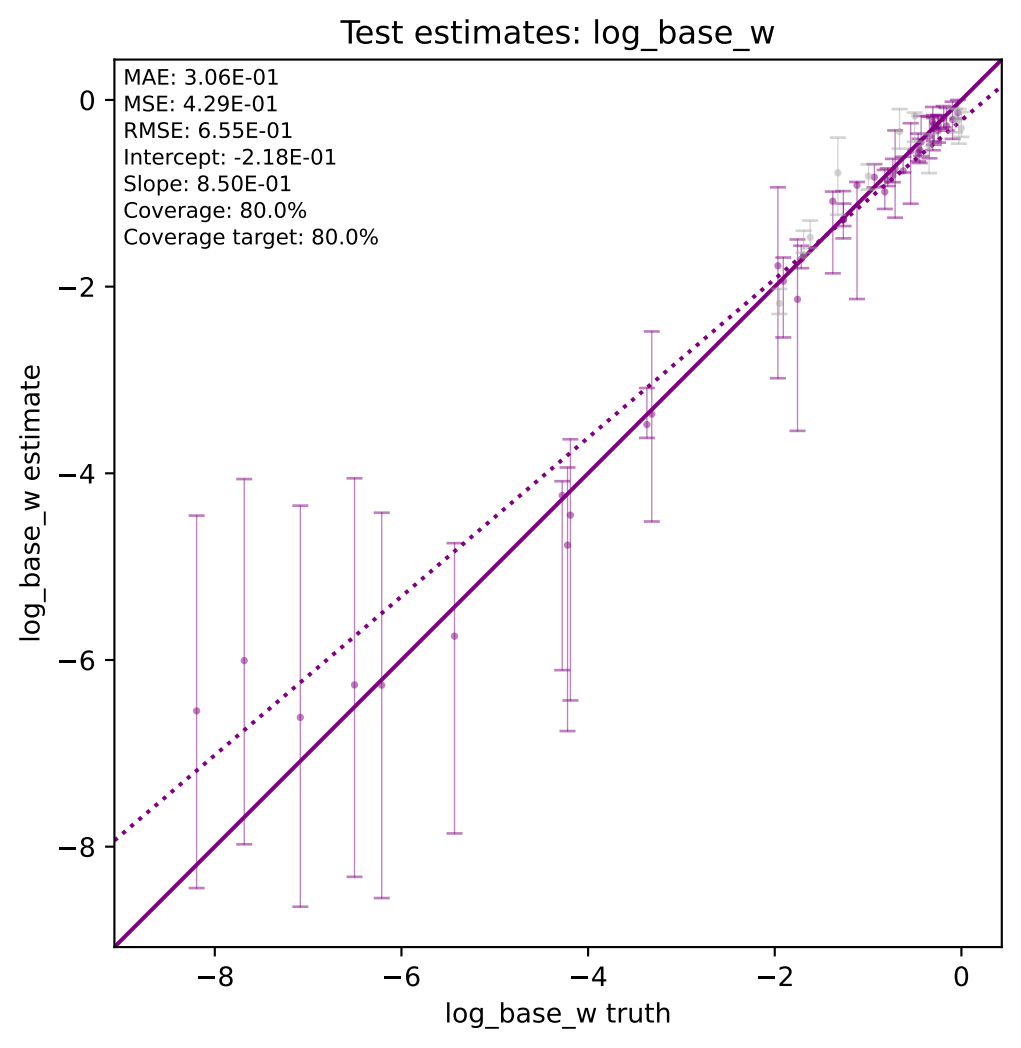

Train estimates: log_base_w
MAE: 2.92E-01
MSE: 4.27E-01
RMSE: 6.54E-01
Intercept: -1.99E-01
Slope: 8.74E-01
Coverage: 81.8%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

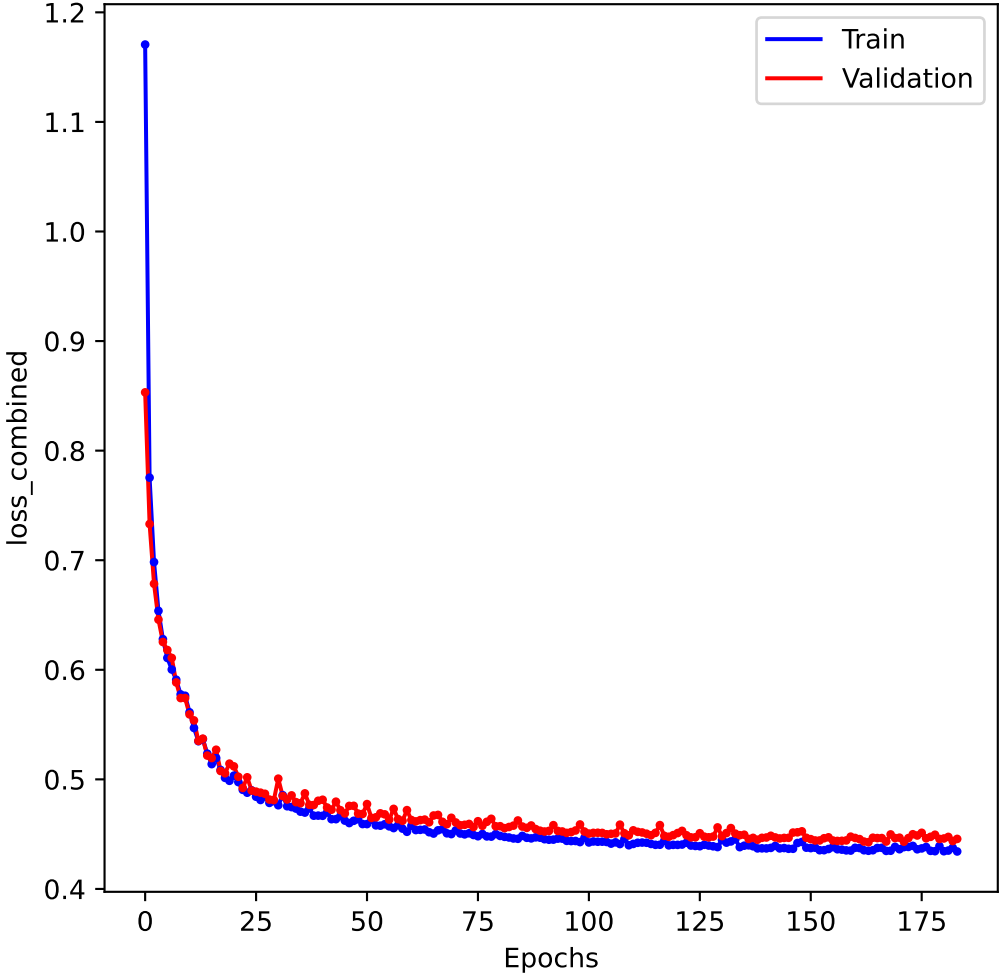

Test estimates: exp_scalar_e
MAE: 1.12E+00
MSE: 2.37E+00
RMSE: 1.54E+00
Intercept: 1.96E+00
Slope: 2.04E-01
Coverage: 80.2%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

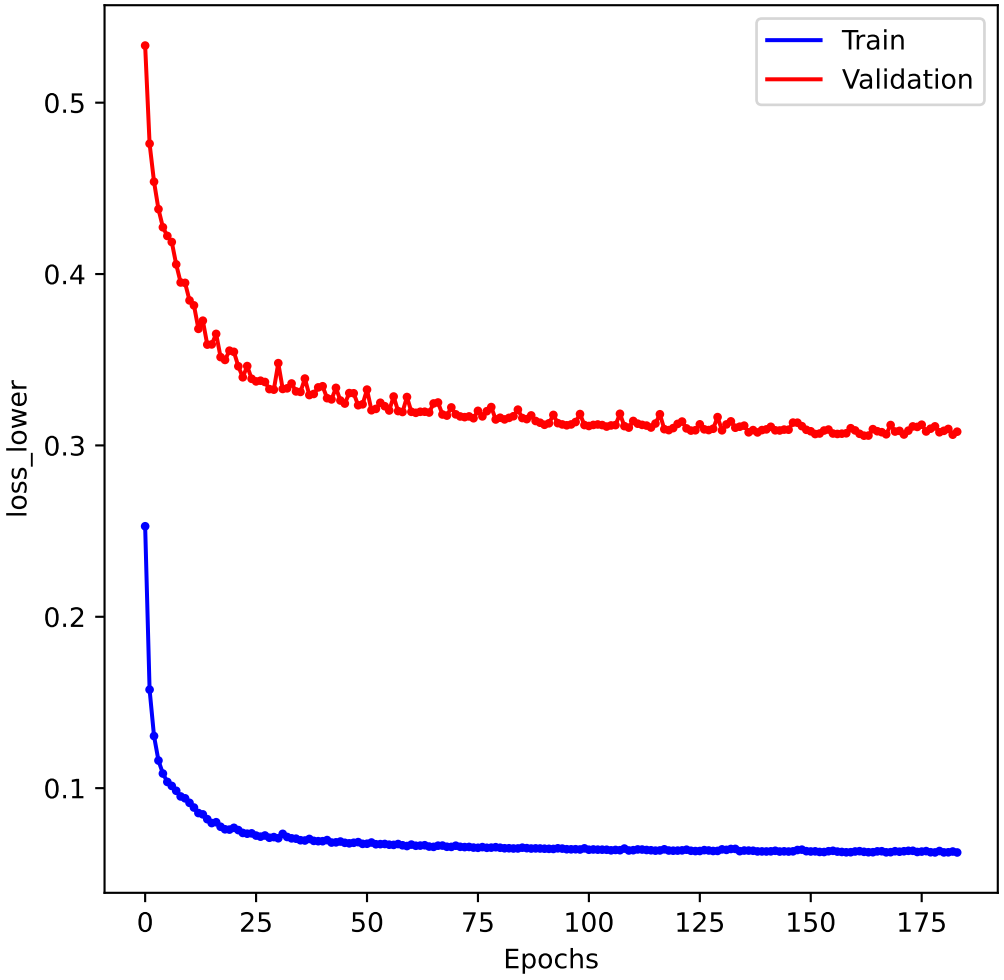

Test estimates: log_base_b
MAE: 8.49E-01
MSE: 1.91E+00
RMSE: 1.38E+00
Intercept: -6.55E-01
Slope: 6.70E-01
Coverage: 80.4%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

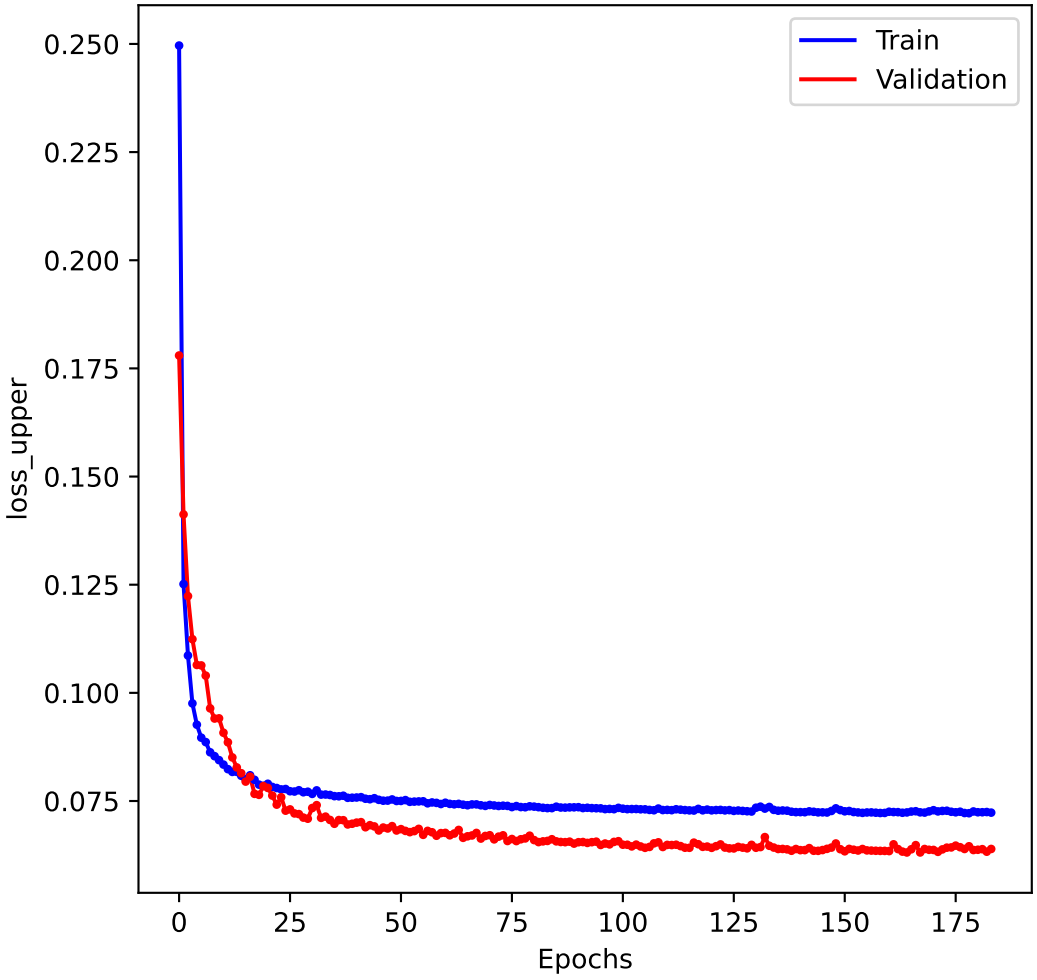

Test estimates: log_base_d
MAE: 2.21E-01
MSE: 1.67E-01
RMSE: 4.09E-01
Intercept: -8.45E-02
Slope: 9.48E-01
Coverage: 81.1%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

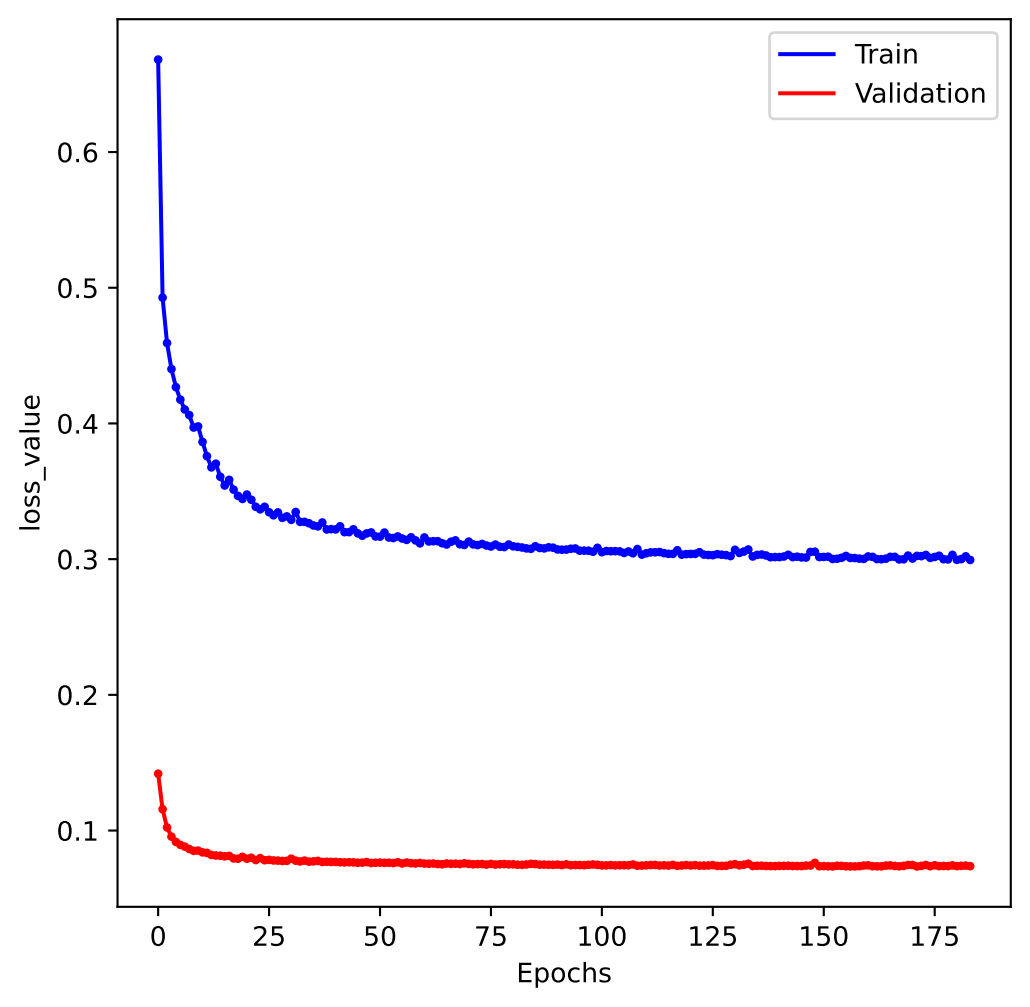

Test estimates: log_base_e
MAE: 6.90E-01
MSE: 8.22E-01
RMSE: 9.07E-01
Intercept: -3.44E-01
Slope: 8.97E-01
Coverage: 80.4%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

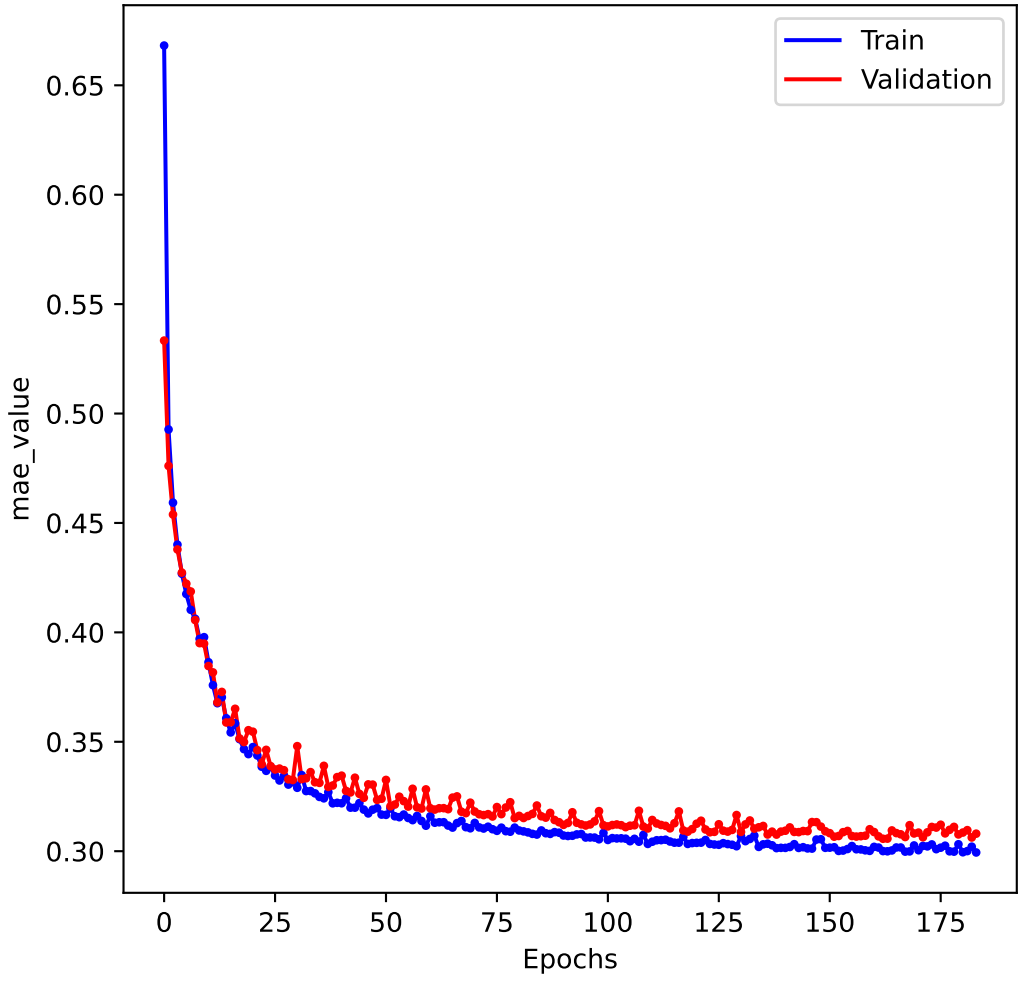

Test estimates: log_base_w
MAE: 3.06E-01
MSE: 4.29E-01
RMSE: 6.55E-01
Intercept: -2.18E-01
Slope: 8.50E-01
Coverage: 80.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

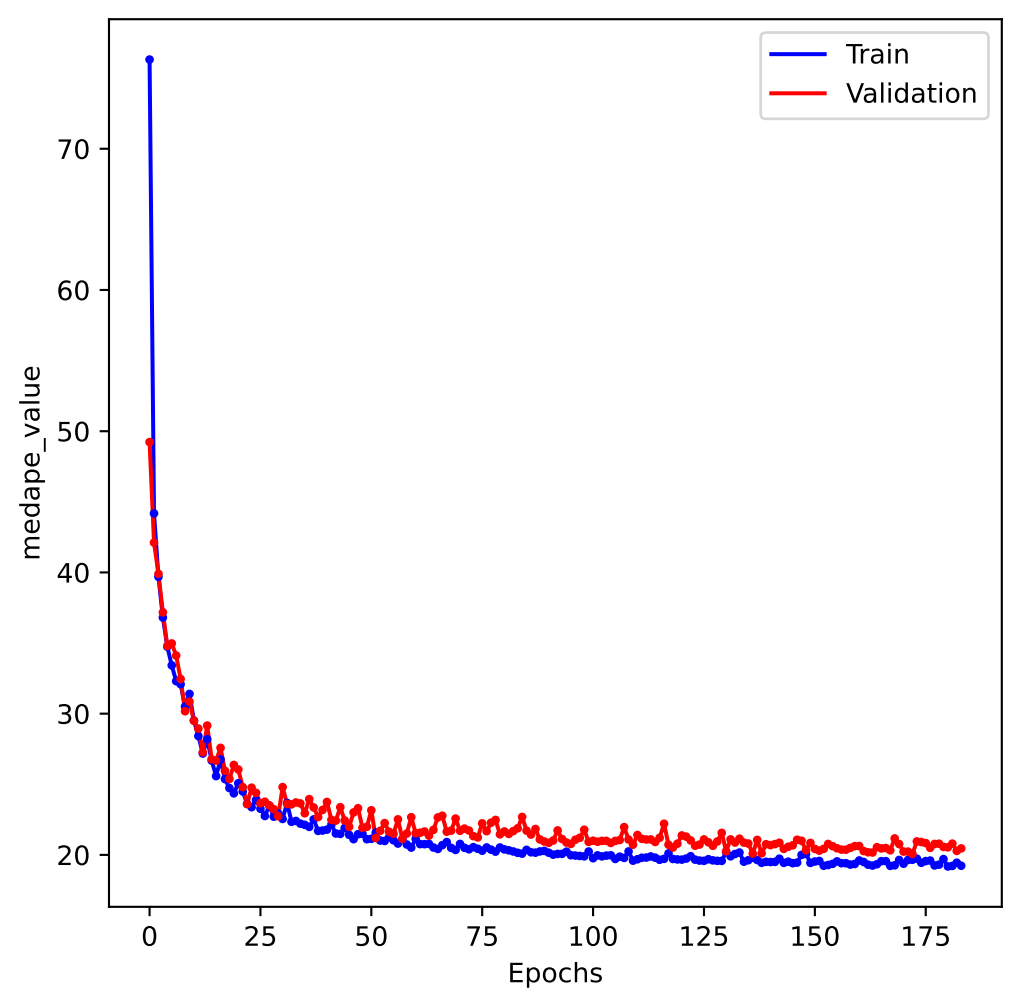

Train
Validation
loss_combined
Epochs
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
25
50
75
100
125
150
175

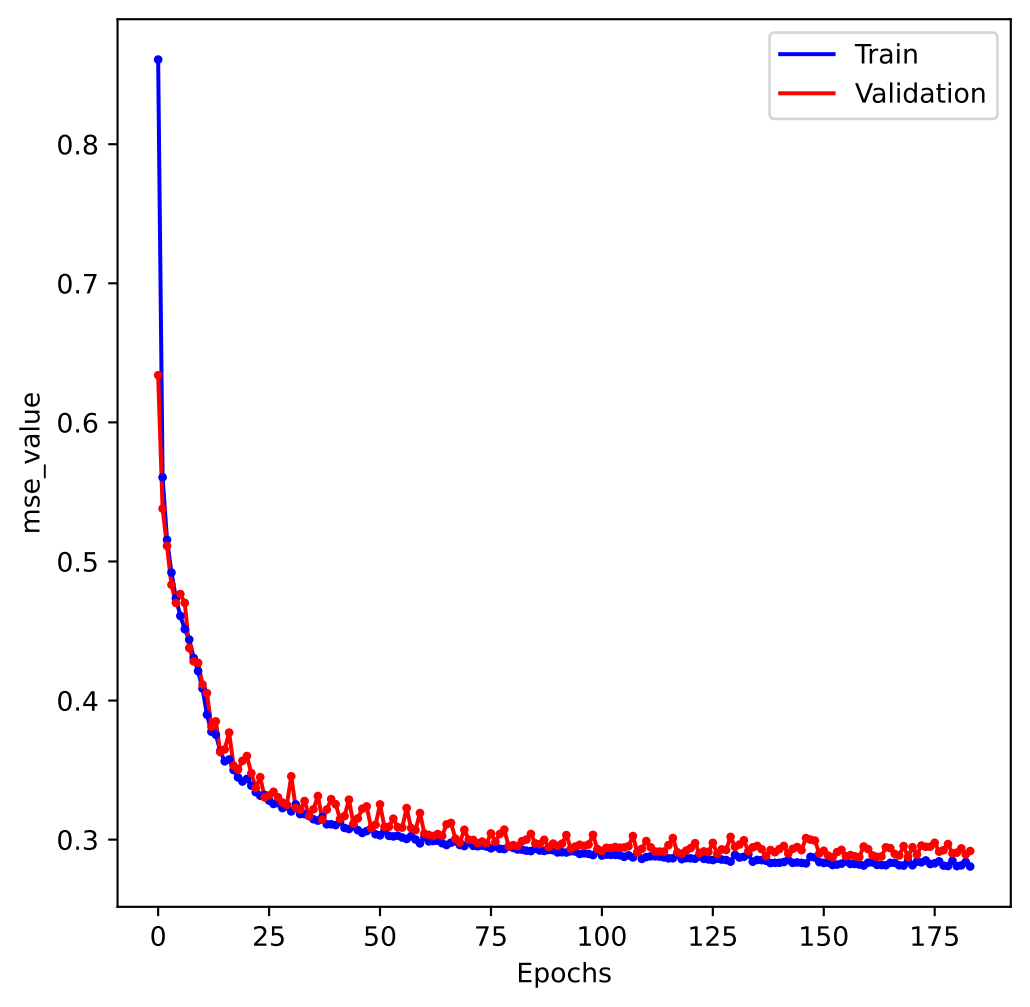

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
25
50
75
100
125
150
175
Epochs

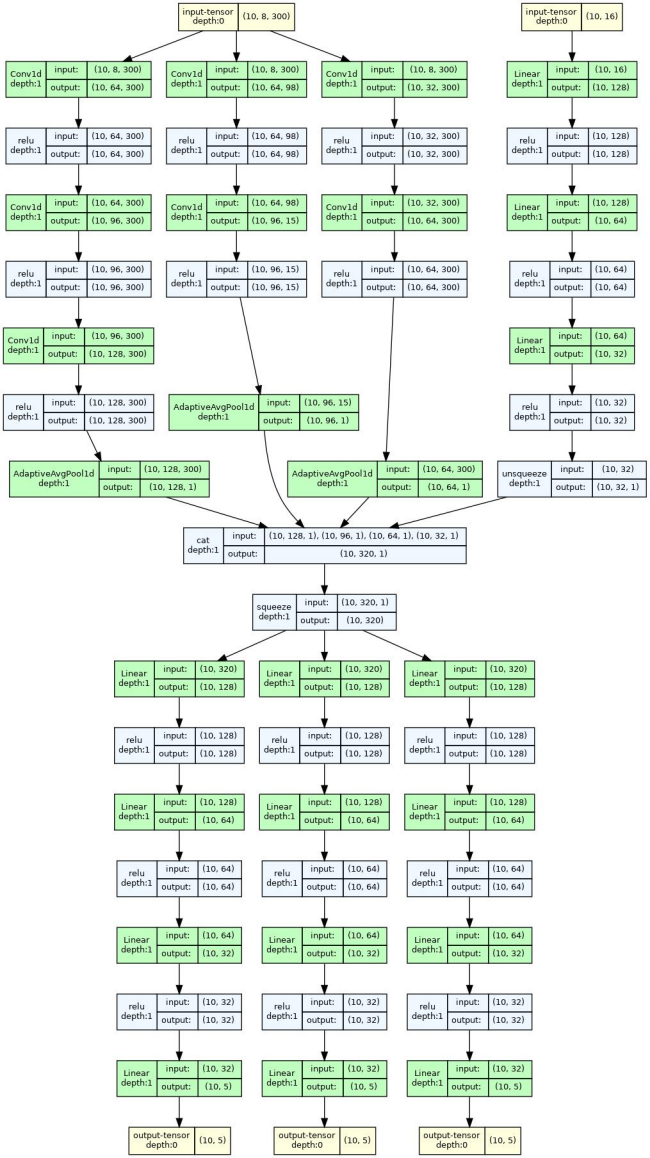

Train
Validation
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
loss_upper
0
25
50
75
100
125
150
175
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
25
50
75
100
125
150
175
Epochs


Train
Validation
mae_value
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0
25
50
75
100
125
150
175
Epochs


Train
Validation
70
60
50
40
30
20
medape_value
0
25
50
75
100
125
150
175
Epochs


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0
25
50
75
100
125
150
175
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)